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Operational Qualification Protocol

1. Process designation

The process to be qualified is the intermediate critical cleaning (see Image A) described in the step 0050 of the manufacturing flowchart FMC187_TUNNELING ROD_CROZET SAS_01E. This flowchart describes the manufacturing process of supplier CROZET SAS and includes the cleaning process with a Dürr Ecoclean EcoCcompact 80-1-L-EU cleaning system (identification number 100050, serial number OK2021005, Image B). This equipment is used to clean components of the cleaning family CFN014 represented by the worst case A1528 BARBED TUNNELING ROD (see §5. below).

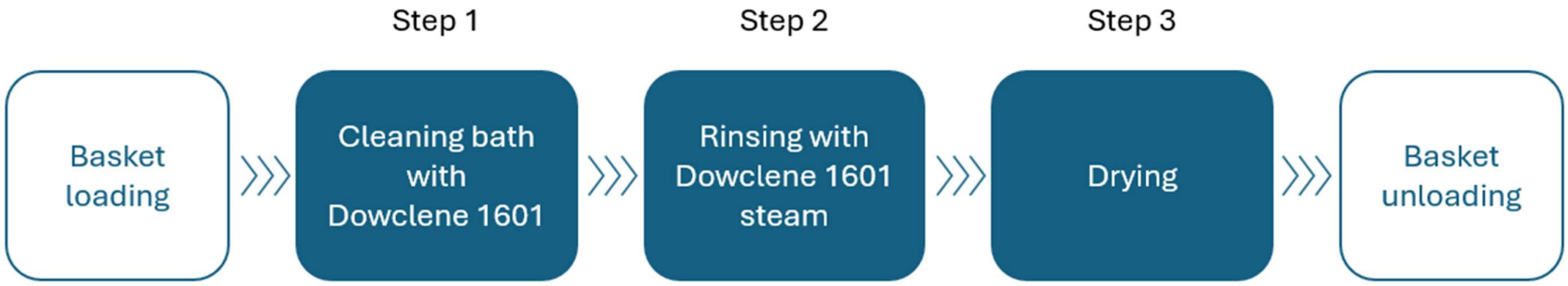


Image A: representative scheme of the intermediate critical cleaning steps

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This equipment cleans by immersion in titanium baskets. The baths (steps 1 and 2, Image A) are carried out at set time and temperature. Depending on the baths, ultrasounds and baskets rotation/oscillation can be added. At the end, a drying chamber allows the evaporation of DOWCLENE 1601 (step 3, Image A). The solvent is filtered at 100 µm before each bath. Hydrocarbon residues are filtered by distillation and released. In addition, for the programs applied to the B.Braun pieces, the movements, temperature, power of ultrasound are fixed and defined.



Image B: photography of identification plate of the Dürr Ecoclean EcoCcompact cleaning system

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2. Origin (Project, CAPA, Change…)

Project PRO0167: Celsite Venous Access Port Systems.

Instruction SA-FR02-M-5-1-06-001-0-R-FR "Validation des procédés"

3. Purpose

The aim of an operational qualification is to establish by objective evidence that the process, under the intended conditions of implementation, consistently produces products in accordance with all pre-established requirements.

Therefore, the purpose of the present validation is to demonstrate that the intermediate critical cleaning process performed by the supplier CROZET SAS with a Dürr Ecoclean EcoCcompact 80-1-L-EU cleaning system (identification number 100050, serial number OK2021005) allows to obtain products in a cleanliness state allowing them to be used in further manufacturing steps according to the NF ISO 19227:2018 standard.

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4. Applicable standards

Standard number	Standard designation	Date
Ph. Eur. <2.2.44>	European Pharmacopoeia (Ph. Eur.)– General chapter 2.2.44 Total Organic Carbon	/
USP <643>	United States Pharmacopeia-National Formulary (USP–NF) General Chapter <643> Total Organic Carbon	/
NF EN ISO 9377-2	Water quality — Determination of hydrocarbon oil index Part 2: Method using solvent extraction and gas chromatography	2000

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5. Components concerned (Worst case rationale when applicable)

Reference	**Main** or alternative designations
A1528	**BARBED TUNNELING ROD**
	Tunneliseur étage
	Tunneliseur

The cleaning family determination is based at least on manufacturing processes, environment, material and adjuvants.

According to the instruction SA-FR02-M-5-1-06-009-0-H-FR "Validation du nettoyage" and the document “CLEANING FAMILY AND WORST CASE.xls” the cleaning family concerned by the present validation is the CFN014 and the component worst case to be tested in the case of validation is the reference A1528 BARBED TUNNELING ROD (specification SPA1528_05). This worst case is representative of the references presented in the table below.

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		Cleaning	
Reference	**Designation**	**Cleaning familly** (for cleaned reference only)	**Worstcase determination index** (Morphological index* Slenderness* Surface)
A1528	**BARBED TUNNELING ROD**	**CFN014**	539346.76

* Reference not included in PRO0167

The worst case determination is based at least on dimensions and morphological criteria (VAP439_01E).

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6. Equipment impacted

Process owner	CROZET SAS
Equipment designation	Dürr Ecoclean EcoCcompact 80-1-L-EU cleaning system
Equipment identification number	100050
Equipment serial number	OK2021005
Tooling identification number (if applicable)	N/A
Tooling serial number (if applicable)	N/A

7. Processing aids

Family Adjuvant Name
004 - SOLVANT ALCOOL MODIFIE

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8. Risk analysis

The risk analysis of the contamination during the manufacturing process is documented in AMD0018_Ed03. The risk analysis allowed to identify intermediate critical cleaning to be validated. Thus, the packaging does not allow to maintain the cleanliness of the pieces. Complementary cleaning is needed before entrance into the controlled manufacturing environment at B.Braun Chasseneuil site. This cleaning will be validated in independent validation.

According to FMC187_01E and MPS411_01E, manufacturing adjuvants are used so the potential contaminants are chemicals. The objective of this intermediate critical cleaning is to remove potential chemicals contaminants. Thus, the chosen analyses to perform in this cleaning validation are HCT and COT

The associated cleanliness criteria are described in the instruction SA-FR02-M-5-1-06-009-0-H-FR "Validation du nettoyage".

9. Batch size identification

According to historical data of the supplier CROZET SAS the usual batch size to be cleaned is 1 000 pieces (corresponding to a total cleaned surface of 2 059 180 mm²) considering minimum batches size of 100 pieces (corresponding to a total cleaned surface of 205 918 mm²) and maximum batches size of 5 000 pieces (corresponding to a total cleaned surface of 10 295 900 mm²).

These data were considered to determine the batches size for this validation.

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10. Environmental conditions (Cleanroom…)

The equipment impacted by the present validation is the Dürr Ecoclean EcoCcompact 80-1-L-EU cleaning system (identification number 100050) installed on Crozet SAS site (296 Rue César Vuarchex 74950 SCIONZIER), in a non-controlled area.

11. Applicable parameters (Pretest analysis…)

a. Pretests results

The justification of the selected parameter range applicable for operational qualification is established based on feedback and experience of the supplier CROZET SAS from the cleaning programs already implemented at the site. The time, temperature and the basket load are the critical parameters identified for this cleaning. The temperature cannot be changed on the equipment, so it is considered as a fixed parameter.

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b. OQ parameters selected values.

Critical parameter	OQ LOW selected value	OQ HIGH selected value
Basket load (piece)	600 (mini)	2800 (maxi)
Cleaning bath duration (sec)	231 (maxi)	189 (mini)
Rinsing duration (sec)	132 (maxi)	108 (mini)
Drying duration (sec)	143 (maxi)	117 (mini)

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12. Tests

a. OQ LOW

Critical parameter	OQ LOW selected value	Really set value
Basket load (piece)	600	Report only
Cleaning bath duration (sec)	231	Report only
Rinsing duration (sec)	132	Report only
Drying duration (sec)	143	Report only

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Test #	Specified requirement	Test method	Sampling baseline	Calculated sampling	Resp.	Acceptance criteria	Results	Status (Pass/Fail)
1	Determination of total hydrocarbons (HCT)	Report only	According to instruction SA-FR02-M-5-1-06-009-0-H-FR "Validation du nettoyage"	3 samples 1 piece per sample **Sampling laboratory request: 1 piece per sample**	ICARE (St Beauzire)	≤0.5 mg/device	Report only	Report only
2	Determination of total organic carbon (COT)	Report only	According to instruction SA-FR02-M-5-1-06-009-0-H-FR "Validation du nettoyage"	3 samples 1 piece per sample **Sampling laboratory request: 1 piece per sample**	ICARE (St Beauzire)	≤0.5 mg/device	Report only	Report only

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b. OQ HIGH

Critical parameter	OQ HIGH selected value	Really set value
Basket load (piece)	2800	Report only
Cleaning bath duration (sec)	189	Report only
Rinsing duration (sec)	108	Report only
Drying duration (sec)	117	Report only

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Test #	Specified requirement	Test method	Sampling baseline	Calculated sampling	Resp.	Acceptance criteria	Results	Status (Pass/Fail)
3	Determination of total hydrocarbons (HCT)	Report only	According to instruction SA-FR02-M-5-1-06-009-0-H-FR "Validation du nettoyage"	3 samples 1 piece per sample **Sampling laboratory request: 1 piece per sample**	ICARE (St Beauzire)	≤0.5 mg/device	Report only	Report only
4	Determination of total organic carbon (COT)	Report only	According to instruction SA-FR02-M-5-1-06-009-0-H-FR "Validation du nettoyage"	3 samples 1 piece per sample **Sampling laboratory request: 1 piece per sample**	ICARE (St Beauzire)	≤0.5 mg/device	Report only	Report only

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13. Glossary

FMC : Flowchart of Manufacturing processes and Controls
CFN : Cleaning Family Number
PVA : Protocole de VAlidation
MPS : Manufacturing Process Specification
AMD : AMDec = Analyse des modes de défaillance, de leurs effets et de leur criticité
OQ : Operational Qualitification
DFP : Demande de Fabrication de Prototype, production, cleaning and/or packaging traceability document, including manufacturing records if applicable (EF, Enregistrement de Fabrication).
CoC : Certificate of Conformity
VAP : VAlidation Plan

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14. Tests results

Test number	Test Report number	Test Report location
1	Report only	Report only
2	Report only	Report only
3	Report only	Report only
4	Report only	Report only

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15. Traceability

Document type	Document identifier	Document location
DFP for OQlow	Report only	Report only
COC for OQlow	Report only	Report only
DFP for OQhigh	Report only	Report only
COC for OQhigh	Report only	Report only

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16. Conclusion

Operational Qualification status	PASS ☐ FAIL ☐

Revision	Description
01	Creation

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Operational Qualification Protocol

Author	Reviewer	Approver
Name: TRAINSON O.	Name: A. DAVID	Name: A. SALMON
Date: 23/02/2026	Date: 23/02/2026	Date: 23.02.26
Visa:	Visa:	Visa:

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Title: PVA0765_OQ_Automated supplier detergent cleaning_CFN014_01 Initiator: Ophelie _ Trainson

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Title: SA-FR02-M Process validation specialist
Date: Monday, 05 October 2026, 13:39 W. Europe Daylight Time
Meaning: Document signed as Author
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UserName: David, Anthony (davianfr)
Title: SA-FR02-M-Industrialization Manager
Date: Tuesday, 06 October 2026, 08:03 W. Europe Daylight Time
Meaning: Approve Document
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